Smart Greenhouse: Dual-Controller Hardware & System Documentation

ESP32 Master Supervisory Node & STM32F103C8T6 Actuator Slave Platform

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1. System Overview & Data Flow Architecture

This technical manual documents the complete hardware connections, circuit schematics, diagnostic flowcharts, sensor calculations, and physical prototype validation for the Dual-Controller Smart Greenhouse. The system separates heavy data-management tasks (Wi-Fi web hosting, LittleFS multi-tier logging, VPD calculation, and Telegram Bot notifications) on an ESP32 Master from physical 12V/5V hardware actuation (PC cooling fan and SG90 vent servo) on an STM32 Blue Pill Slave connected via an asynchronous 115200 bps UART bridge.

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========================================================== 🌐 USER INTERFACE
LAYER ============================================================ [ Telegram App
] [ Offline Web Dashboard ] ^ ^ | (HTTPS / Wi-Fi) | (Local Wi-Fi) v v
========================================================== 🧠 ESP32 MASTER
CONTROLLER ============================================================ [ INPUTS
] [ OUTPUTS ] ▶ DHT22 Sensor (GPIO 4) ----------------------> ▶ OLED Display (I2C)
▶ BH1750 Light (I2C) ----------------------> ▶ Water Pump Relay (GPIO 26) ▶ Soil
Moisture (GPIO 34) ▶ Push Button (GPIO 13) | | [ INTER-BOARD COMMUNICATION ] | ▶
Transmits: Sensor telemetry | ▶ Transmits: <FAN_ON/OFF> | ▶ Transmits:
<VENT_OPEN/CLOSE> | v (UART: ESP32 TX2 -> STM32 RX1) ^ (UART: ESP32 RX2 <- STM32
TX1) | ========================================================== ⚙️ STM32
SLAVE CONTROLLER (Blue Pill)
========================================================== [ INPUTS ] [ OUTPUTS
] ▶ UART Commands (PA10) ----------------------> ▶ 12V PC Fan via MOSFET (PA0) ---
------------------> ▶ SG90 Servo Vent (PA6)
========================================================== ⚡ POWER
DISTRIBUTION ============================================================ [ 12V
DC Wall Adapter ] ├──> Direct 12V Line ----------------------> 12V PC Fan └──> [ 5V
Buck Converter ] ├──> 5V Line --------------------> STM32, Servo, Relay, ESP32 VIN
└──> [ 3.3V Regulator ] └──> 3.3V Line ---------> Sensors (DHT22, BH1750), OLED
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2. Dual-Controller System Block Diagram

The architectural block diagram below illustrates the complete signal flow between the environmental sensors, the ESP32 Master controller, the I2C display, the Wi-Fi / Telegram cloud interfaces, the UART serial link, and the STM32 Slave actuator stage.

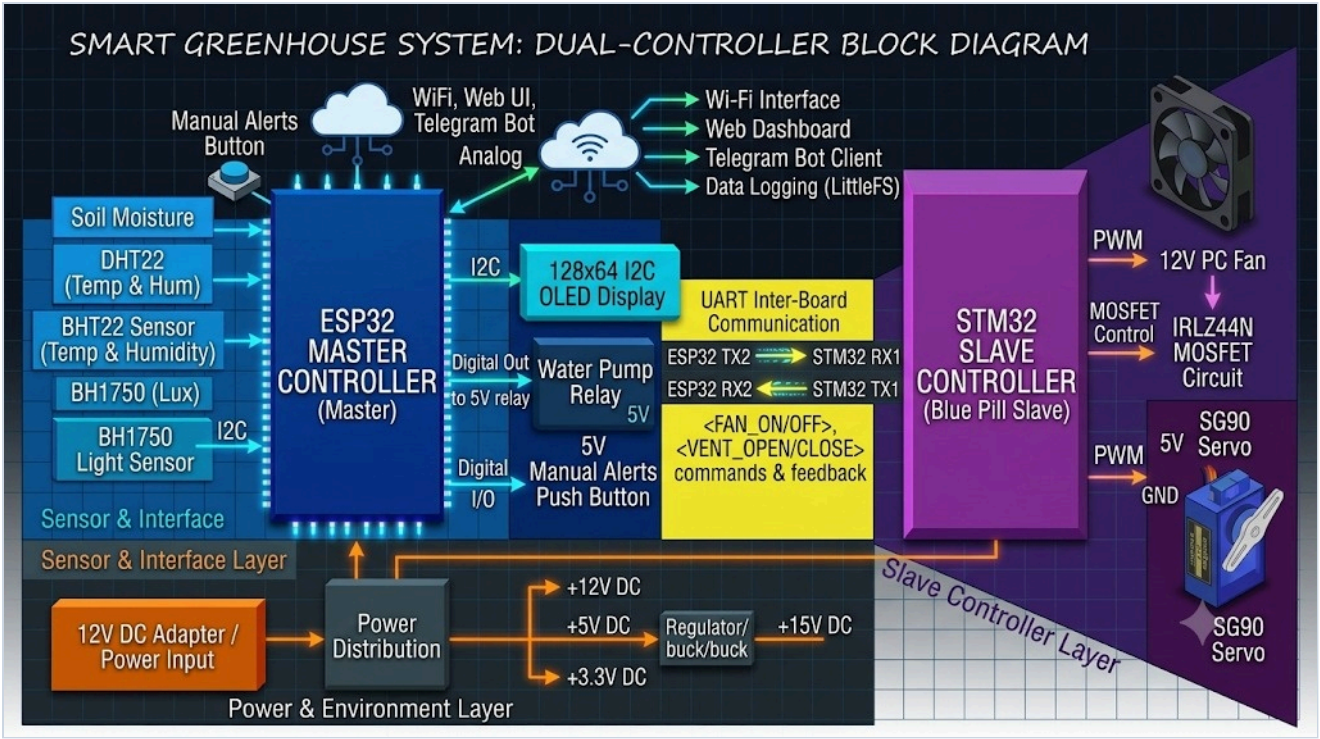


Figure 1: Complete Dual-Controller System Block Diagram showing Sensor/Interface Layer, Power Distribution, and STM32 Slave Layer.

3. Complete Pin-to-Pin Hardware Connections Table

All physical wiring connections across the ESP32 Dev Module (30-pin), STM32F103C8T6 Blue Pill, environmental sensors, MOSFET switching stage, and power rails are detailed in Table 1.

Category	Component / Module	Controller & Pin	Signal Type & Electrical Notes
Sensors	DHT22 (Temperature & RH)	ESP32 — GPIO 4	Digital 1-Wire Data (3.3V supply, 10 kΩ pull-up resistor)
Sensors	BH1750 (Ambient Light)	ESP32 — GPIO 21 (SDA), GPIO 22 (SCL)	I2C Bus (Continuous High-Res Mode, 3.3V logic)
Sensors	Capacitive Soil Moisture Probe	ESP32 — GPIO 34 (ADC1)	Analog ADC input (11 dB attenuation, 16-sample averaging)
Local UI	SSD1306 128×64 OLED	ESP32 — GPIO 21 (SDA), GPIO 22 (SCL)	Shared I2C Bus (Address 0x3C, 3.3V power)
Local UI	Manual Alerts Push Button	ESP32 — GPIO 13	Digital Input (Active LOW to GND, internal pull-up enabled)
Local UI	Status Indicator LED	ESP32 — GPIO 2	Digital Output (Blinks when soil moisture drops below threshold)
Actuators (ESP32)	5V Water Pump Relay	ESP32 — GPIO 26	Digital Output (Active HIGH, 5s auto-run, 60s manual safety limit)

Category	Component / Module	Controller & Pin	Signal Type & Electrical Notes
UART Bridge	Master TX2 → Slave RX1	ESP32 GPIO 17 → STM32 PA10	115200 bps, 8N1 format: sends telemetry & bracketed commands
UART Bridge	Master RX2 ← Slave TX1	ESP32 GPIO 16 ← STM32 PA9	115200 bps crossover serial return line
Actuators (STM32)	12V PC Cooling Fan	STM32 — PA0 (MOSFET Gate)	Drives IRLZ44N Gate via resistor; switches 12V DC low-side path
Actuators (STM32)	SG90 Micro Servo (Roof Vent)	STM32 — PA6 (PWM)	5V power rail; sweeps 0° (Closed) to 90° (Open)
Power Rails	Common System Ground	ESP32 GND ↔ STM32 GND ↔ 12V GND	CRITICAL: All power supplies and boards must share common GND

4. Electrical Schematic & Breadboard Wiring Layout

Figure 2 presents the engineering circuit schematic, including the 1N4007 flyback protection diode across the 12V fan and the IRLZ44N N-channel logic-level MOSFET gate network. Figure 3 shows the corresponding physical breadboard wiring guide.

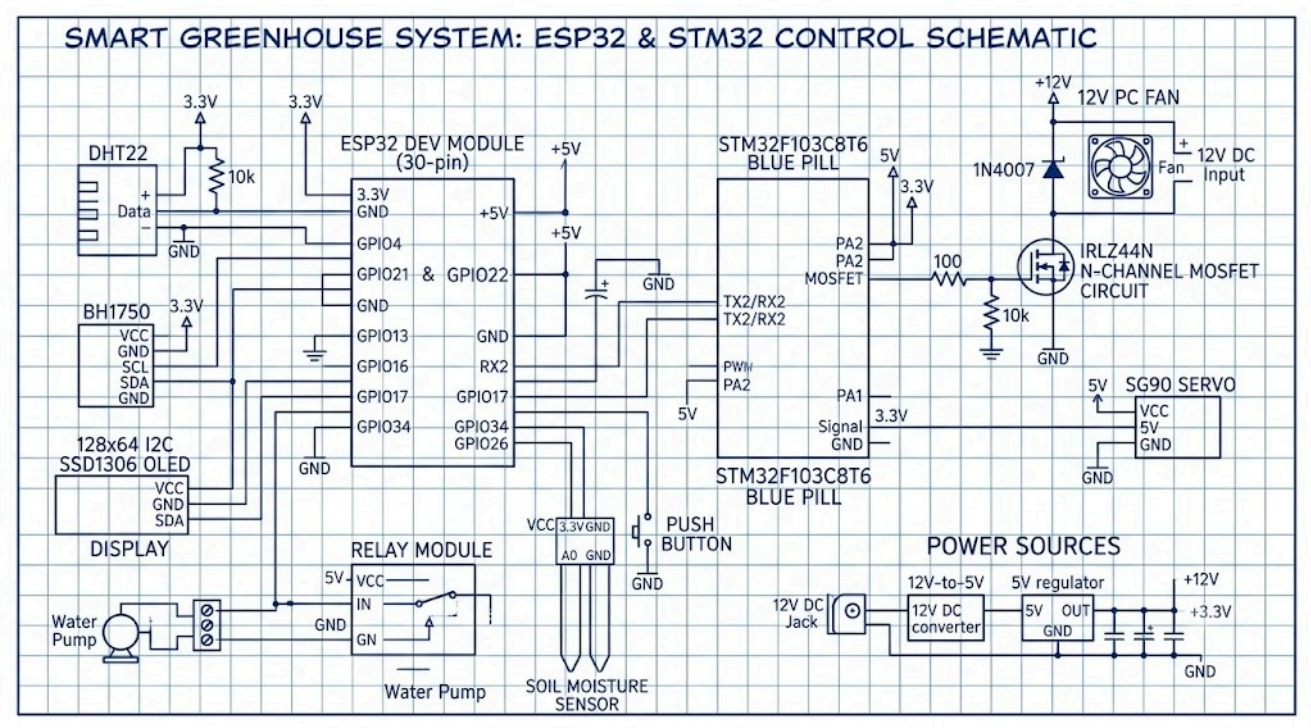


Figure 2: Electrical Schematic Diagram detailing ESP32, STM32, sensor buses, relay driver, and IRLZ44N MOSFET fan stage.

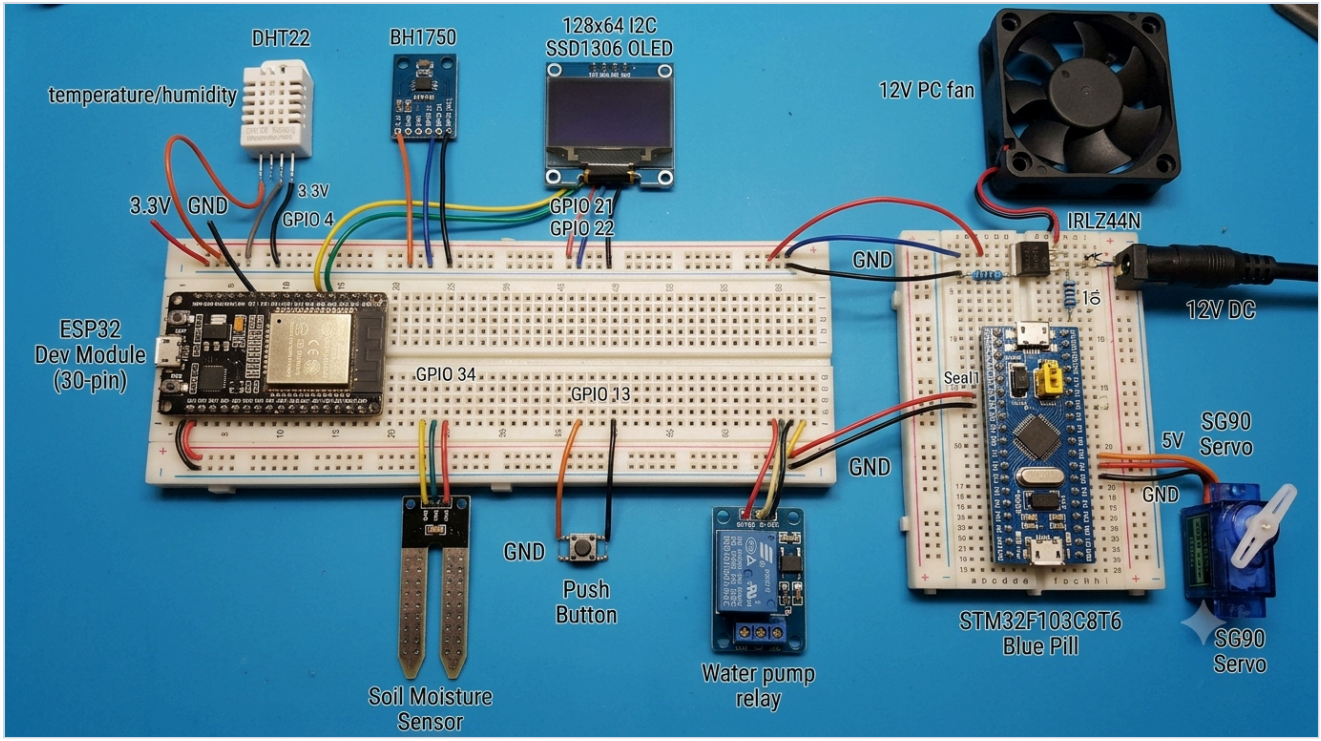


Figure 3: Complete Workbench Breadboard Wiring Diagram showing exact jumper connections between ESP32, STM32, sensors, and actuators.

5. Sensor Signal Processing & Mathematical Formulas

To eliminate analog noise and prevent rapid actuator chattering, the ESP32 firmware processes raw readings every 2000 ms using the following equations:

> **1. Capacitive Soil Moisture EMA Filter & Calibration Mapping:**
> $m_f[n] = 0.3 \times m_{raw}[n] + 0.7 \times m_f[n-1]$
> Moisture (%) = ((Air_Val − $m_f[n]$) ÷ (Air_Val − Water_Val)) × 100 (Default: Air_Val = 3500, Water_Val = 1500)

> **2. DHT22 Temperature & Humidity EMA Smoothing:**
> $T_f[n] = 0.2 \times T_{raw}[n] + 0.8 \times T_f[n-1]$ | $H_f[n] = 0.2 \times H_{raw}[n] + 0.8 \times H_f[n-1]$

> **3. Vapor Pressure Deficit (VPD) Agronomic Stress Calculation:**
> SVP = 0.6108 × exp((17.27 × T) ÷ (T + 237.3)) | VPD (kPa) = SVP × (1 − (RH ÷ 100))

> **4. UART Telemetry Frame & Additive Checksum:**
> Packet Format: <Moisture, Lux, Temp, Humidity, Checksum>
> Checksum = int(Moisture) + int(Lux) + int(Temp) + int(Humidity)

6. Hardware Diagnostic & Verification Flowchart

Figure 4 illustrates the sequential 5-stage diagnostic workflow used to verify voltage rails, I2C address acknowledgment, web UI synchronization, UART packet transmission, and STM32 actuator execution.

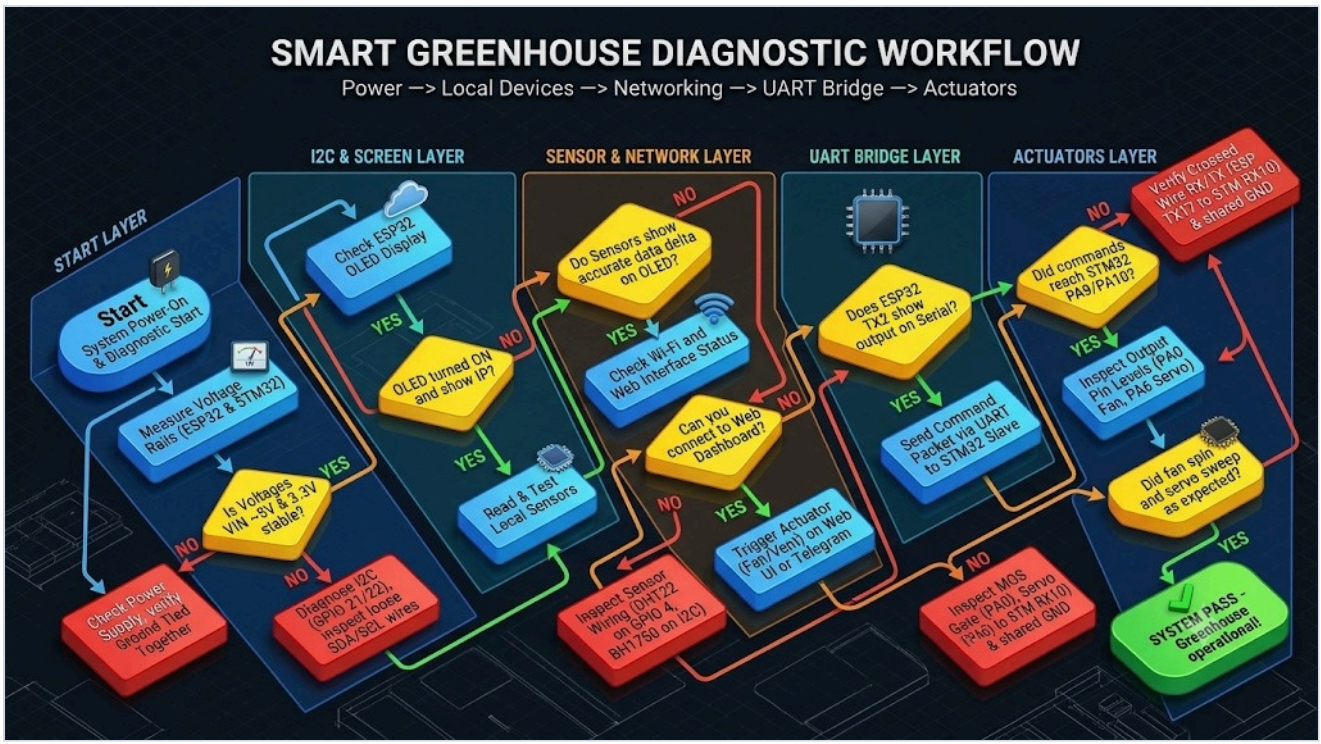


Figure 4: Sequential Hardware Diagnostic Flowchart (Start Layer → I2C/Screen → Sensor/Network → UART Bridge → Actuators).

7. Assembled Physical Prototype & Circuit Photographs

The photographs below document the completed physical greenhouse prototype, including the green shade-net enclosure, live plant substrate with the capacitive soil probe and DHT22 sensor, and the dual-controller breadboard assembly.

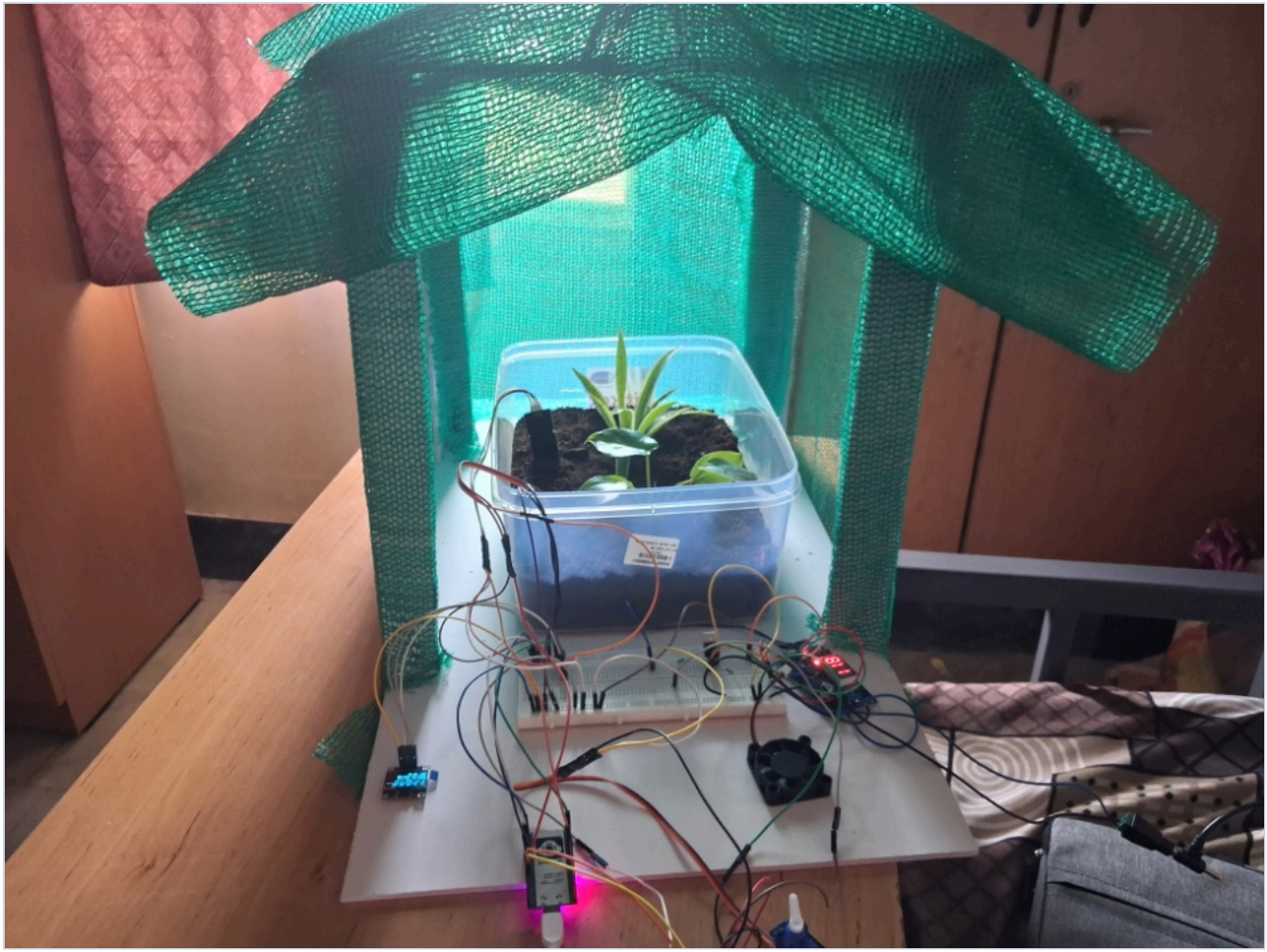


Figure 5: Front view of the assembled Smart Greenhouse prototype showing the shade-net structure, plant container, OLED display, 12V fan, and wiring.

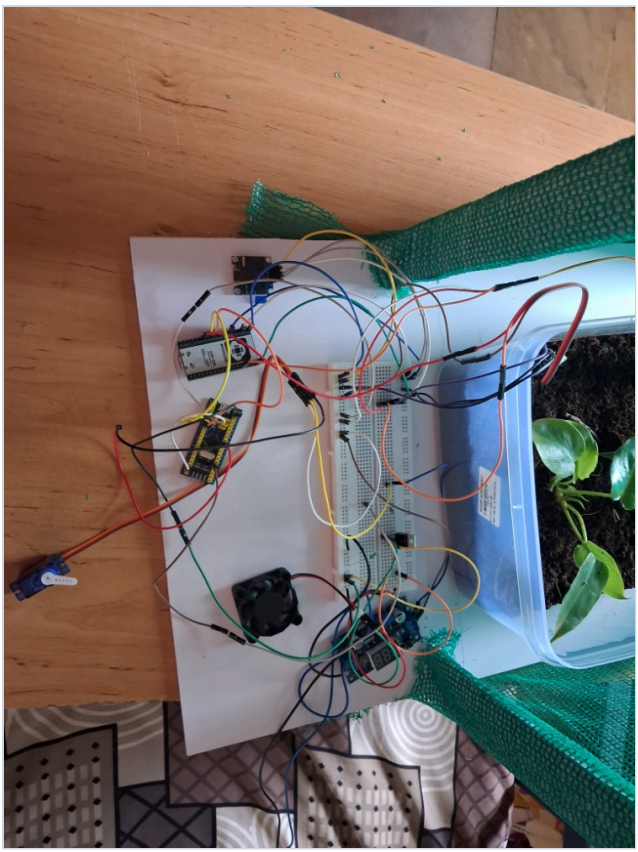


Figure 6: Top-down view of the hardware baseboard showing the ESP32 Master, STM32 Blue Pill Slave, 12V cooling fan, SG90 servo, and buck converter.



Figure 7: Interior view of the greenhouse plant bed housing the test plants in soil substrate.

8. Live Web Dashboard & Multi-Tier Telemetry Results

During live hardware testing (IP: `10.185.105.157`), the ESP32 web server accurately rendered real-time environmental metrics (28.4 °C temperature, 84.1%–84.2% air humidity, 0.61–0.62 kPa VPD, and 48/100 Plant Health Score), interactive actuator toggles, LittleFS historical plots, and automated watering event logs.

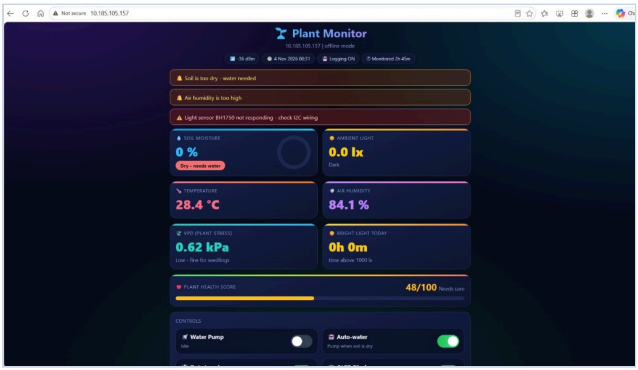


Figure 8: Live Web Dashboard displaying active alert banners, 28.4 °C temperature, 84.1% humidity, and 0.62 kPa VPD.

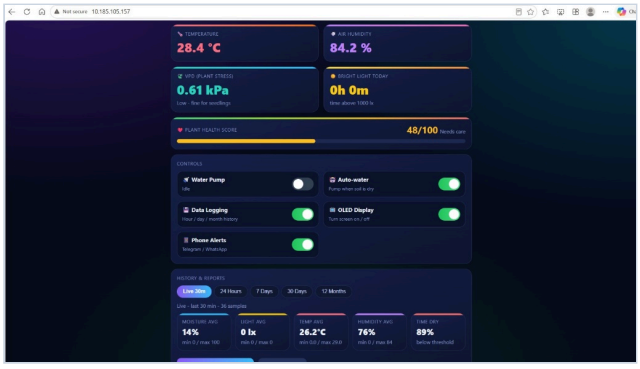


Figure 9: Dashboard Control Panel showing toggles for Water Pump, Auto-water, Data Logging, OLED Display, and Phone Alerts alongside 30-min averages.

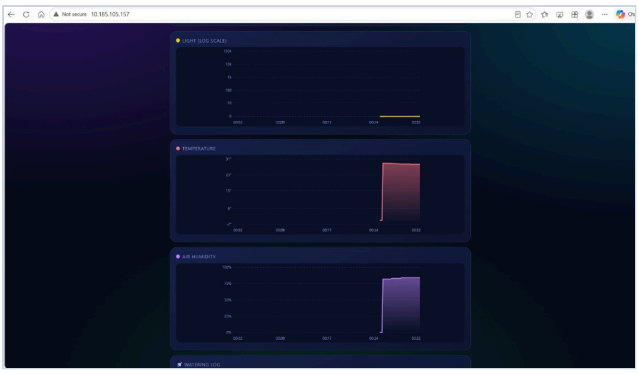


Figure 10: Real-time browser-rendered SVG trend charts for Ambient Light (log scale), Temperature, and Air Humidity.

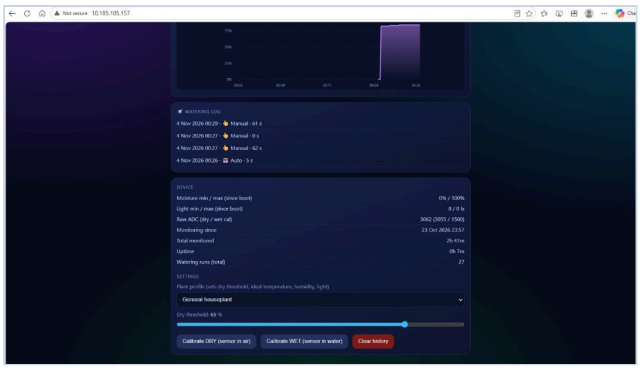


Figure 11: Watering Event Log (27 total runs), raw ADC calibration readout (3062 raw), and Plant Profile selector.